

BISHAL SAMANTA

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RESEARCH INTERESTS

Application of machine learning and deep learning to diverse biological data modalities, including microscopy images, genomic sequences, transcriptomics, and molecular structures. Interested in developing and adapting modern deep learning approaches — representation learning, self-supervised learning, graph neural networks, and transformers — to extract meaningful biological signal from complex, high-dimensional datasets. Motivated by problems where computational methods can directly accelerate biological discovery, from quantitative morphological phenotyping and bioimage analysis to genomic variant interpretation and multi-omics data integration

EDUCATION

University of Göttingen Göttingen, Germany
Master of Science in Computational Biology and Bioinformatics 2024 – 2026

GPA: 1.3/4.0 (German grading scale)

Relevant Coursework: Computational Methods in Biology, Data Analytics, Molecular Modeling, Advanced Bioinformatics

National Institute of Technology, Durgapur Durgapur, India
Bachelor of Technology in Biotechnology 2020 – 2024

GPA: 4.0/4.0 — *First Class with Distinction*

Relevant Coursework: Genetics, Molecular Biology, Programming for Bioinformatics, Biostatistics, Machine Learning

TECHNICAL SKILLS

Programming & Computing: Python, R, SQL, Java, C; Git, MATLAB, Jupyter Notebook, Anaconda
Machine Learning & AI: Deep Learning, Neural Networks, Model Interpretability (SHAP), Hyperparameter Tuning, Cross-validation, Ensemble Methods, Transformers, Language Models for Biological Sequences, VAEs, Graph Networks

Bioinformatics & Structural Biology: AlphaFold, RosettaFoldNA, DRFold2, ViennaRNA/RNAfold, SimRNA, Protein Structure Prediction, RNA 3D Structure Prediction, Multi-omics Integration

Data Analysis & Visualization: Pandas, NumPy, Seaborn, Plotly, Statistical Analysis, Large-scale Genomic Data Analysis, Transcriptome Analysis, BLAST, ExPASy, Gene Ontology, Pathway Enrichment Analysis

Scientific Computing: Pipeline Development, Computational Workflows, Parallel Computing, High-throughput Analysis

Bioimage Analysis: 3D Microscopy Image Processing, Expansion Microscopy (U-ExM), Representation Learning for Morphology, Unsupervised Feature Extraction, Self-supervised Learning, PyTorch, TensorFlow

RESEARCH EXPERIENCE

Scientific Trainee (*Theory@EMBL Fellow*) May 2026 – Present
European Molecular Biology Laboratory (EMBL) Heidelberg, Germany

- Developing an **unsupervised representation learning framework for 3D morphological analysis** of ultrastructure expansion microscopy (U-ExM) data, eliminating the need for manual segmentation or labeled training data
- Designing **modality-agnostic deep learning architectures** (VAEs, self-supervised models) to automatically extract morphological features directly from 3D images, enabling universally applicable analysis across diverse imaging methods and biological contexts
- Investigating **centrin structural diversity in free-living unicellular eukaryotes** distributed across the tree of life, leveraging learned representations to characterize and compare ultrastructural phenotypes at scale

- Contributing a generalizable computational tool for **quantitative cellular morphology analysis** to the broader biomedical research community, with the goal of streamlining the study of structure–function relationships in cell biology

Student Research Assistant (HiWi)

Max Planck Institute for Multidisciplinary Sciences

Jan 2025 – Apr 2026

Göttingen, Germany

Supervisor: Dr. Peter Lenart

- Engineered automated **protein structure prediction pipeline using AlphaFold** to evaluate functional impacts of transcriptomic changes at the protein level, processing large-scale datasets with 95%+ accuracy
- Performed comprehensive **computational analysis of transcriptomic data** to identify 3' UTR regulatory regions during oocyte maturation, integrating motif discovery, RNA secondary structure prediction, and pathway enrichment analysis
- Investigated 3' UTR features through **motif identification and gene ontology mapping** to uncover conserved regulatory elements linked to key cellular processes
- Developed **computational workflows for multi-omics integration**, combining transcriptome data with structural biology to elucidate molecular mechanisms underlying developmental biology

ERASMUS+ Research Intern

International Institute of Molecular and Cell Biology (IIMCB)

Oct 2025 – Dec 2025

Warsaw, Poland

- Built **meta-server architecture for RNA 3D structure prediction** by integrating multiple deep learning models (RosettaFoldNA, DRFold2), improving prediction accuracy through ensemble approaches
- Designed and implemented **Python-based pipeline scripts** to merge multi-tool predictions and generate composite structural inputs for SimRNA modeling, demonstrating strong software engineering skills
- Optimized computational workflows for **high-throughput structural predictions**, reducing processing time by 40% through parallel computing strategies and efficient resource management

Bioinformatics Research Intern

University of Delhi

May 2023 – Jul 2023

New Delhi, India

Supervisor: Dr. Manish Kumar

- Developed **production-ready machine learning pipeline in Python** for antibiotic resistance prediction with 90%+ accuracy, incorporating advanced data visualization using Seaborn and Plotly
- Implemented **SHAP (SHapley Additive exPlanations) for model interpretability**, identifying key genomic features driving resistance predictions and improving clinical applicability of ML models
- **Published first-author research in peer-reviewed journal** (*Bioinformatics Advances*, Oxford Academic) and presented findings at international conference, demonstrating strong scientific communication skills
- Collaborated in cross-disciplinary research team, contributing to interpretable machine learning methods for clinical bioinformatics applications

RESEARCH PROJECTS

Machine Learning-Based Classification of Allergenic Proteins

National Institute of Technology, Durgapur

Nov 2023 – Apr 2024

Durgapur, India

Supervisor: Dr. Amita Barik

- Curated balanced allergenic and non-allergenic protein datasets from COMPARE-2024, Allergen Online, and AlgPred 2.0, ensuring data quality through CD-HIT clustering and refinement
- Analyzed **motif patterns and physicochemical properties** including amino acid composition and secondary structure to identify discriminative features of allergenic proteins
- Developed **ensemble voting classifier** optimized through hyperparameter tuning and 10-fold cross-validation, achieving robust performance measured by accuracy, precision, recall, and ROC-AUC metrics

Computational Prediction of Small Non-coding RNAs in *Salmonella enterica*

Oct 2023

Aug 2023 –

National Institute of Technology, Durgapur

Durgapur, India

Supervisor: Dr. Amita Barik

- Obtained and curated **large-scale sRNA datasets** from open-access bioinformatics repositories, ensuring comprehensive coverage and data quality for downstream analysis

- Performed CD-HIT clustering and **feature extraction using Vienna RNA package** to analyze structural and functional characteristics of sRNA molecules
- Developed **machine learning predictive model** for sRNA-related biological functions through structured feature transformation and model training
- **Presented poster at International Conference** on Global Disease Biology (University of Burdwan, 2024), receiving valuable feedback from international scientific community

PUBLICATIONS

Jain, A., et al. (2025). An explainable machine learning pipeline for prediction of antimicrobial resistance. *Bioinformatics Advances*, Oxford Academic. DOI: [10.1093/bioadv/vbaf190](https://doi.org/10.1093/bioadv/vbaf190) (Second Author)

CONFERENCE PRESENTATIONS

Computational prediction of small non-coding RNAs in *Salmonella enterica* — Poster Presentation

International Conference on Global Disease Biology: Perspective Biotechnology, University of Burdwan, 2024

ADVANCED TRAINING & CERTIFICATIONS

Summer School in Applied Deep Learning in Bioinformatics *August 2025*
University of Copenhagen — Text Mining, Transformers, Language Models for Biological Sequences, Variational Autoencoders, Graph Neural Networks

Bioinformatics and Biostatistics Certification *June 2023*
Association for Computing Activities, Department of Computer Science & Engineering, IIT Kanpur

NextGen Helper ENSEMBL Genome Browser and REST API Workshop *Oct 2022*
EMBL-EBI — Virtual Workshop

HONOURS & AWARDS

Theory@EMBL Fellowship *2026*
European Molecular Biology Laboratory — Competitive fellowship supporting theoretical and computational research at EMBL Heidelberg

PROFESSIONAL ACTIVITIES

Volunteer — International Society of Computational Biology (ISCB), Regional Student Group India
Active participant in computational biology community, contributing to knowledge sharing and scientific outreach

ADDITIONAL SKILLS

Strong analytical and problem-solving abilities with meticulous attention to detail • Effective collaboration and communication in cross-disciplinary research teams • Proven project management skills with ability to deliver results under deadlines • Scientific rigor and commitment to reproducible research